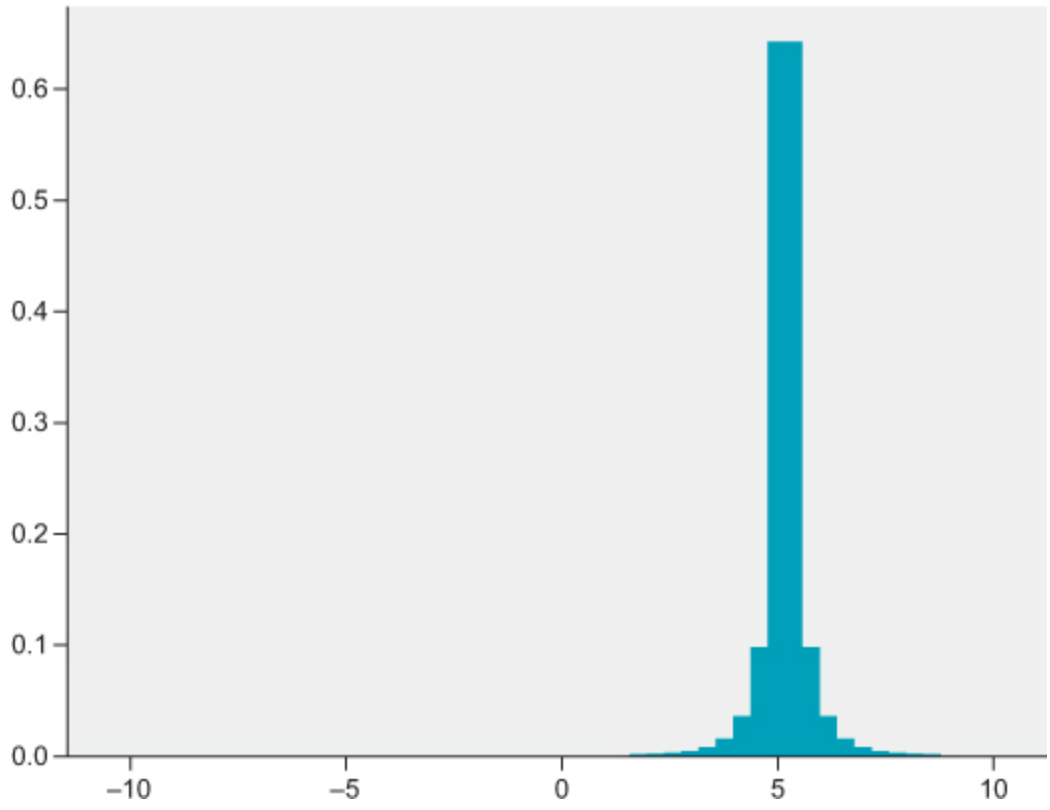


7.4.2. Implementing the Dist-DQN

As we briefly discussed earlier, the Dist-DQN will take a 128-element state vector, pass it through a couple of dense feedforward layers, and then it will use a `for` loop to multiply the last layer by 3 separate matrices to get 3 separate distribution vectors. We will lastly apply the softmax function to ensure it is a valid probability distribution. The result is a neural network with 3 different output "heads." We collect these 3 output distributions into a single 3 × 51 matrix and return that as the final output of the Dist-DQN. Thus, we can get the individual action-value distributions for a particular action by indexing a particular row of the output matrix. Figure 7.15 shows the overall architecture and tensor transformations. In listing 7.6 we define the function that implements the Dist-DQN.

Figure 7.15. The Dist-DQN accepts a 128-element state vector and produces 3 separate 51-element probability distribution vectors, which then get stacked into a single 3 × 51 matrix.